Lesson 54: Policy Gradient Theorem

AI/ML Mathematics

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1 Motivation and Overview

Consider a robot learning to navigate a continuous environment. The robot must choose actions from a continuous action space – for instance, joint torques or wheel velocities – based on high-dimensional sensory input. A value-based approach would require computing $\arg\max_a Q(s, a)$ over this continuous space at every step, which is computationally intractable without further structure. Instead, we would like to directly parameterize a policy π_θ – a mapping from states to distributions over actions – and optimize the parameters θ by gradient ascent on the expected cumulative reward. But how do we compute the gradient of an expectation that depends on θ through both the action selection and the resulting state distribution? The policy gradient theorem, developed in this lesson, provides the answer: a remarkably clean expression for $\nabla_\theta J(\theta)$ that avoids differentiating through the state transition dynamics.

RL/ML Connection

- **REINFORCE** [Williams, 1992]: Uses the policy gradient theorem with sampled returns to form a Monte Carlo gradient estimator.
- **Actor-critic methods** [Konda and Tsitsiklis, 2003]: Replace the action-value function in the policy gradient with a learned approximation (the critic), reducing variance.
- **Trust region methods** [Schulman et al., 2015]: Build on the policy gradient by constraining updates to a KL-divergence ball for stable optimization.
- **Natural policy gradient** [Kakade, 2002]: Preconditions the policy gradient with the inverse Fisher information matrix, using the score function structure derived here.

Prerequisites. The reader should be familiar with: Markov decision processes (states, actions, transition kernel, reward function, discount factor γ), value functions $V^\pi(s)$ and $Q^\pi(s, a)$, and the Bellman equations (Lessons 22–23). From probability theory, we require conditional expectation and the law of total expectation (Lesson 11), convergence of series and interchange of summation/differentiation under uniform convergence (Lesson 4, Theorem 3.2). Basic multivariate calculus including gradients and the chain rule (Lesson 3) is assumed throughout.

Learning objectives.

- **State** the log-derivative trick and explain why it converts a gradient of an expectation into an expectation of a gradient.
- **Derive** the policy gradient theorem for the episodic (start-state) setting from first principles.
- **Prove** the policy gradient theorem for the average-reward setting.
- **Prove** that subtracting a state-dependent baseline from the action-value function does not bias the policy gradient.
- **Compute** score functions for softmax and Gaussian policy parameterizations.
- **Explain** the compatible function approximation condition and its role in actor-critic architectures.

2 References

- Sutton and Barto [2018, Chapter 13] – Policy gradient methods with intuitive derivations and algorithmic descriptions.
- Sutton et al. [2000] – The original policy gradient theorem paper; Theorem 1 and the compatible function approximation result.
- Williams [1992] – REINFORCE and the likelihood ratio method for gradient estimation.
- Peters and Schaal [2008] – Natural actor-critic, connecting policy gradients to information geometry.
- Bertsekas and Tsitsiklis [1996, Chapter 6] – Gradient-based methods in neuro-dynamic programming, convergence analysis.
- Kakade [2002] – Natural policy gradient derivation and compatible features.

3 Parameterized Policies and the Performance Objective

Throughout this lesson, we work with a Markov decision process $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$ where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space (finite unless stated otherwise), $\mathcal{P}(s' \mid s, a)$ is the transition kernel, $r(s, a)$ is the immediate reward bounded by $|r(s, a)| \leq R_{\max}$, and $\gamma \in [0, 1)$ is the discount factor.

Definition 3.1 (Parameterized Policy)**.** A *parameterized policy* is a family of stochastic policies $\{\pi_\theta : \theta \in \Theta \subseteq \mathbb{R}^d\}$, where for each θ, $\pi_\theta(a \mid s)$ defines a probability distribution over $\mathcal{A}$ for every state $s \in \mathcal{S}$. We assume that $\pi_\theta(a \mid s) > 0$ for all s, a, θ and that $\pi_\theta(a \mid s)$ is differentiable with respect to θ.

Definition 3.2 (Performance Objective – Start-State Formulation)**.** Given a fixed start state s_0 (or a distribution μ over start states), the *discounted start-state objective* is

$$J(\theta) = V^\pi[\theta](s_0) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t\, r(S_t, A_t) \;\middle|\; S_0 = s_0\right],$$

where the expectation is taken over trajectories generated by following π_θ. When a start distribution μ is used, we write $J(\theta) = \mathbb{E}_{s_0 \sim \mu}[V^\pi[\theta](s_0)] = \sum_s \mu(s)\, V^\pi[\theta](s)$.

Definition 3.3 (Discounted State Visitation Distribution)**.** The *discounted state visitation distribution* under policy π_θ, starting from state s_0 (or distribution μ), is

$$d^{\pi_\theta}(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t\, \mathbb{P}(S_t = s \mid S_0 \sim \mu,\, \pi_\theta).$$

The factor $(1 - \gamma)$ normalizes d^{π_θ} to a proper probability distribution: $\sum_s d^{\pi_\theta}(s) = 1$. When the context is clear, we write d^π omitting the subscript θ.

Remark 3.4. Some references (including Sutton et al. [2000]) use the *unnormalized* visitation distribution $\tilde{d}^\pi(s) = \sum_{t=0}^{\infty} \gamma^t\, \mathbb{P}(S_t = s \mid s_0, \pi)$, which sums to $1/(1-\gamma)$. We will state the policy gradient theorem using $\tilde{d}^\pi$ to match the standard form, noting that the normalization constant does not affect gradient ascent since it is θ-independent.

Definition 3.5 (Average-Reward Objective)**.** For continuing tasks, the *average-reward objective* is

$$J_{\text{avg}}(\theta) = \lim_{T\to\infty} \frac{1}{T}\mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} r(S_t, A_t)\right] = \sum_s d_\infty^{\pi_\theta}(s) \sum_a \pi_\theta(a \mid s)\, r(s,a),$$

where $d_\infty^{\pi_\theta}$ is the stationary distribution of the Markov chain induced by π_θ (assumed to exist and be unique).

Example 3.6 (Softmax Policy)**.** For finite $\mathcal{A} = \{a_1, \dots, a_k\}$, define features $\phi(s,a) \in \mathbb{R}^d$ and let

$$\pi_\theta(a \mid s) = \frac{\exp(\theta^\top \phi(s,a))}{\sum_{a'} \exp(\theta^\top \phi(s,a'))}.$$

This is the *softmax policy*. It satisfies $\pi_\theta(a \mid s) > 0$ for all a, s, θ, so the differentiability assumption in Definition 3.1 holds.

Example 3.7 (Gaussian Policy)**.** For continuous $\mathcal{A} \subseteq \mathbb{R}^m$, a common choice is the *Gaussian policy* with state-dependent mean:

$$\pi_\theta(a \mid s) = \frac{1}{(2\pi\sigma^2)^{m/2}} \exp\left(-\frac{\|a - \mu_\theta(s)\|^2}{2\sigma^2}\right),$$

where $\mu_\theta(s) = \theta^\top \phi(s)$ is a linear function of state features and $\sigma^2 > 0$ is a fixed variance. Here $\theta \in \mathbb{R}^{d\times m}$ and $\phi(s) \in \mathbb{R}^d$.

We now turn to the fundamental identity that makes policy gradient methods possible: the log-derivative trick.

4 The Log-Derivative Trick

The central challenge in computing $\nabla_\theta J(\theta)$ is that the expectation is taken with respect to a distribution that depends on θ. The log-derivative trick (also called the likelihood ratio trick or score function method) provides a way to move the gradient inside the expectation.

Lemma 4.1 (Score Function Identity)**.** *Let $p_\theta(x)$ be a parameterized probability distribution over x with $p_\theta(x) > 0$ for all x in the support, and let $f(x)$ be a function independent of θ. Suppose that differentiation and summation (or integration) can be interchanged. Then*

$$\nabla_\theta \mathbb{E}_{x\sim p_\theta}[f(x)] = \mathbb{E}_{x\sim p_\theta}\big[f(x)\, \nabla_\theta \log p_\theta(x)\big].$$

Proof. For the discrete case (the continuous case is identical with integrals replacing sums):

$$\begin{aligned}
\nabla_\theta \mathbb{E}_{x\sim p_\theta}[f(x)] &= \nabla_\theta \sum_x p_\theta(x)\, f(x) = \sum_x f(x)\, \nabla_\theta p_\theta(x) \\
&= \sum_x f(x)\, p_\theta(x)\, \frac{\nabla_\theta p_\theta(x)}{p_\theta(x)} = \sum_x p_\theta(x)\, f(x)\, \nabla_\theta \log p_\theta(x) \\
&= \mathbb{E}_{x\sim p_\theta}\big[f(x)\, \nabla_\theta \log p_\theta(x)\big].
\end{aligned}$$

The key step uses the identity $\nabla_\theta \log p_\theta(x) = \nabla_\theta p_\theta(x)/p_\theta(x)$, which follows from the chain rule applied to $\log(\cdot)$. □

Lemma 4.2 (Score Function Has Zero Mean)**.** *Under the same conditions as Lemma 4.1,*

$$\mathbb{E}_{x\sim p_\theta}\big[\nabla_\theta \log p_\theta(x)\big] = 0.$$

Proof. Apply Lemma 4.1 with $f(x) = 1$:

$$\mathbb{E}_{x\sim p_\theta}\big[\nabla_\theta \log p_\theta(x)\big] = \nabla_\theta \mathbb{E}_{x\sim p_\theta}[1] = \nabla_\theta 1 = 0.$$ □

The Log-Derivative Trick

For a parameterized policy $\pi_\theta(a \mid s)$:

$$\nabla_\theta \, \pi_\theta(a \mid s) = \pi_\theta(a \mid s) \, \nabla_\theta \log \pi_\theta(a \mid s).$$

This identity transforms the gradient of a probability (which may be hard to sample from) into an expectation under that same probability, enabling Monte Carlo estimation.

Example 4.3 (Score Function for Softmax Policy)**.** Recall the softmax policy from Example 3.6. We have $\log \pi_\theta(a \mid s) = \theta^\top \phi(s,a) - \log \sum_{a'} \exp(\theta^\top \phi(s,a'))$, so

$$\nabla_\theta \log \pi_\theta(a \mid s) = \phi(s,a) - \sum_{a'} \pi_\theta(a' \mid s) \, \phi(s,a') = \phi(s,a) - \mathbb{E}_{a' \sim \pi_\theta(\cdot \mid s)}[\phi(s,a')].$$

The score function is the feature vector of the chosen action minus the expected feature vector under the current policy.

Example 4.4 (Score Function for Gaussian Policy)**.** For the Gaussian policy of Example 3.7 with mean $\mu_\theta(s) = \theta^\top \phi(s)$ and fixed variance $\sigma^2 I$,

$$\log \pi_\theta(a \mid s) = -\frac{\|a - \theta^\top \phi(s)\|^2}{2\sigma^2} + \text{const},$$

so the score function is

$$\nabla_\theta \log \pi_\theta(a \mid s) = \frac{(a - \theta^\top \phi(s)) \, \phi(s)^\top}{\sigma^2}.$$

For scalar actions and parameters, this simplifies to $\nabla_\theta \log \pi_\theta(a \mid s) = \frac{(a - \theta^\top \phi(s))}{\sigma^2} \, \phi(s)$. The score is proportional to the "surprise" $(a - \mu_\theta(s))$ weighted by the feature vector.

With the log-derivative trick established, we are now ready to derive the policy gradient theorem.

5 The Policy Gradient Theorem (Episodic Setting)

The policy gradient theorem expresses $\nabla_\theta J(\theta)$ in a form that does not require differentiating the state transition dynamics $\mathcal{P}(s' \mid s,a)$, which may be unknown or complex.

Theorem 5.1 (Policy Gradient Theorem – Episodic Setting)**.** *Let $J(\theta) = V^\pi[\theta](s_0)$ be the discounted start-state objective. Under the differentiability and interchange conditions of Definition 3.1, we have*

$$\nabla_\theta J(\theta) = \sum_{s \in \mathcal{S}} \tilde{d}^{\pi_\theta}(s) \sum_{a \in \mathcal{A}} \nabla_\theta \, \pi_\theta(a \mid s) \, Q^\pi[\theta](s,a),$$

where $\tilde{d}^{\pi_\theta}(s) = \sum_{t=0}^{\infty} \gamma^t \, \mathbb{P}(S_t = s \mid s_0, \pi_\theta)$ is the unnormalized discounted state visitation. Equivalently, using the log-derivative trick,

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim \tilde{d}^{\pi_\theta}, \, a \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s) \, Q^\pi[\theta](s,a)\big].$$

Proof. We proceed by deriving a recursive formula for $\nabla_\theta V^\pi(s)$ and then unrolling the recursion.

Step 1: Gradient of the value function. Starting from the Bellman equation $V^\pi(s) = \sum_a \pi_\theta(a \mid s) \, Q^\pi(s,a)$, we take the gradient with respect to θ:

$$\nabla_\theta V^\pi(s) = \sum_a \big[\nabla_\theta \pi_\theta(a \mid s) \, Q^\pi(s,a) + \pi_\theta(a \mid s) \, \nabla_\theta Q^\pi(s,a)\big]. \tag{1}$$

Step 2: Expanding $\nabla_\theta Q^\pi(s,a)$. Since $Q^\pi(s,a) = r(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s,a)\, V^\pi(s')$ and neither r nor $\mathcal{P}$ depend on θ,

$$\nabla_\theta Q^\pi(s,a) = \gamma \sum_{s'} \mathcal{P}(s' \mid s,a)\, \nabla_\theta V^\pi(s'). \tag{2}$$

Step 3: Substitution and recursion. Substituting (2) into (1):

$$\begin{aligned}\nabla_\theta V^\pi(s) &= \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a) + \gamma \sum_a \pi_\theta(a \mid s) \sum_{s'} \mathcal{P}(s' \mid s,a)\, \nabla_\theta V^\pi(s') \\ &= \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a) + \gamma \sum_{s'} \mathbb{P}(S_1 = s' \mid S_0 = s, \pi_\theta)\, \nabla_\theta V^\pi(s'). \end{aligned} \tag{3}$$

Step 4: Unrolling the recursion. Define $\psi(s) = \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a)$. Then (3) reads $\nabla_\theta V^\pi(s) = \psi(s) + \gamma \sum_{s'} \mathbb{P}^\pi(s \to s', 1)\, \nabla_\theta V^\pi(s')$, where $\mathbb{P}^\pi(s \to s', k)$ denotes the probability of reaching s' from s in k steps under π_θ. Unrolling:

$$\begin{aligned}\nabla_\theta V^\pi(s) &= \psi(s) + \gamma \sum_{s'} \mathbb{P}^\pi(s \to s', 1) \left[\psi(s') + \gamma \sum_{s''} \mathbb{P}^\pi(s' \to s'', 1)\, \nabla_\theta V^\pi(s'')\right] \\ &= \psi(s) + \gamma \sum_{s'} \mathbb{P}^\pi(s \to s', 1)\, \psi(s') + \gamma^2 \sum_{s''} \mathbb{P}^\pi(s \to s'', 2)\, \nabla_\theta V^\pi(s'') \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{s'} \mathbb{P}^\pi(s \to s', t)\, \psi(s'). \end{aligned}$$

The series converges because $|\psi(s')| \le \|\nabla_\theta \pi_\theta\|_1 \cdot R_{\max}/(1-\gamma)$ is bounded, and $\sum_{s'} \mathbb{P}^\pi(s \to s', t) = 1$.

Step 5: Evaluating at $s = s_0$. Setting $s = s_0$:

$$\begin{aligned}\nabla_\theta J(\theta) = \nabla_\theta V^\pi(s_0) &= \sum_{t=0}^{\infty} \gamma^t \sum_s \mathbb{P}(S_t = s \mid S_0 = s_0, \pi_\theta) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a) \\ &= \sum_s \underbrace{\left(\sum_{t=0}^{\infty} \gamma^t\, \mathbb{P}(S_t = s \mid s_0, \pi_\theta)\right)}_{\tilde{d}^{\pi_\theta}(s)} \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a). \end{aligned}$$

Step 6: Log-derivative form. Applying the identity $\nabla_\theta \pi_\theta(a \mid s) = \pi_\theta(a \mid s)\, \nabla_\theta \log \pi_\theta(a \mid s)$ to the inner sum:

$$\sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a) = \sum_a \pi_\theta(a \mid s)\, \nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi(s,a) = \mathbb{E}_{a \sim \pi_\theta(\cdot \mid s)}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi(s,a)\big].$$

Combining with the outer sum yields the expectation form. □

Policy Gradient Theorem (Sutton et al., 2000)

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi[\theta](s,a)\big].$$

The gradient of the performance objective equals the expected product of the *score function* $\nabla_\theta \log \pi_\theta(a \mid s)$ and the *action-value function* $Q^\pi(s,a)$, weighted by the on-policy state distribution. Crucially, the gradient of the transition dynamics $\mathcal{P}$ does not appear.

RL/ML Connection

The policy gradient theorem is the foundation of all policy-based RL algorithms. The key practical insight is that we can *estimate* the gradient from sample trajectories: generate episodes under π_θ, compute (or estimate) $Q^\pi(s_t, a_t)$ for visited state-action pairs, and average the products $\nabla_\theta \log \pi_\theta(a_t \mid s_t) \cdot Q^\pi(s_t, a_t)$ to obtain a stochastic gradient. This is exactly what REINFORCE and actor-critic algorithms do.

Model-Free, Not Assumption-Free

The policy gradient theorem does not require knowledge of $\mathcal{P}$, but it does require that $\pi_\theta(a \mid s) > 0$ for all s, a (so that $\log \pi_\theta$ is well-defined) and that the interchange of differentiation and summation is valid. In practice, deterministic policies ($\pi_\theta(a \mid s) \in \{0, 1\}$) require a separate treatment (see Lesson 58 on deterministic policy gradients).

6 The Policy Gradient Theorem (Average-Reward Setting)

For continuing tasks without discounting, we use the average-reward objective $J_{\text{avg}}(\theta)$ from Definition 3.5.

Definition 6.1 (Differential Value Function)**.** The *differential value function* under π_θ is

$$V^{\pi_\theta}_{\text{diff}}(s) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty}\big(r(S_t, A_t) - J_{\text{avg}}(\theta)\big) \;\middle|\; S_0 = s\right],$$

and the *differential action-value function* is

$$Q^{\pi_\theta}_{\text{diff}}(s, a) = r(s, a) - J_{\text{avg}}(\theta) + \sum_{s'} \mathcal{P}(s' \mid s, a)\, V^{\pi_\theta}_{\text{diff}}(s').$$

These satisfy the differential Bellman equation: $V^{\pi}_{\text{diff}}(s) = \sum_a \pi_\theta(a \mid s)\, Q^{\pi}_{\text{diff}}(s, a)$.

Assumption 6.2. The Markov chain induced by π_θ is ergodic for all θ: it is irreducible and aperiodic, so a unique stationary distribution $d^{\pi_\theta}_\infty$ exists.

Theorem 6.3 (Policy Gradient Theorem – Average-Reward Setting)**.** *Under Assumption 6.2, the gradient of the average-reward objective is*

$$\nabla_\theta J_{\text{avg}}(\theta) = \sum_s d^{\pi_\theta}_\infty(s) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^{\pi_\theta}_{\text{diff}}(s, a).$$

Equivalently,

$$\nabla_\theta J_{\text{avg}}(\theta) = \mathbb{E}_{s \sim d^{\pi_\theta}_\infty,\, a \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, Q^{\pi_\theta}_{\text{diff}}(s, a)\big].$$

Proof. We differentiate the stationarity condition and the average-reward identity.

Step 1. The average reward satisfies $J_{\text{avg}}(\theta) = \sum_s d^{\pi_\theta}_\infty(s) \sum_a \pi_\theta(a \mid s)\, r(s, a)$. Differentiating:

$$\nabla_\theta J_{\text{avg}} = \sum_s \big[\nabla_\theta d^{\pi_\theta}_\infty(s)\big] \sum_a \pi_\theta(a \mid s)\, r(s, a) + \sum_s d^{\pi_\theta}_\infty(s) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, r(s, a). \tag{4}$$

Step 2. Differentiate the differential Bellman equation $V^{\pi}_{\text{diff}}(s) = \sum_a \pi_\theta(a \mid s)\, Q^{\pi}_{\text{diff}}(s, a)$ with respect to θ:

$$\nabla_\theta V^{\pi}_{\text{diff}}(s) = \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^{\pi}_{\text{diff}}(s, a) + \sum_a \pi_\theta(a \mid s)\, \nabla_\theta Q^{\pi}_{\text{diff}}(s, a).$$

Since $\nabla_\theta Q^\pi_{\text{diff}}(s,a) = -\nabla_\theta J_{\text{avg}} + \sum_{s'} \mathcal{P}(s' \mid s,a)\, \nabla_\theta V^\pi_{\text{diff}}(s')$, multiply by $d^\pi_\infty(s)$, sum over s, and use stationarity:

$$\sum_s d^\pi_\infty(s)\, \nabla_\theta V^\pi_{\text{diff}}(s) = \sum_s d^\pi_\infty(s) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi_{\text{diff}}(s,a) - \nabla_\theta J_{\text{avg}} + \sum_{s'} d^\pi_\infty(s')\, \nabla_\theta V^\pi_{\text{diff}}(s').$$

The left side and last term on the right cancel, yielding the result.

The log-derivative form follows by the same argument as in Theorem 5.1. □

7 Baseline Invariance

A key property of the policy gradient is that we can subtract any state-dependent function from Q^π without introducing bias. This will be essential for variance reduction in Lesson 55.

Proposition 7.1 (Baseline Invariance). *For any function $b : \mathcal{S} \to \mathbb{R}$,*

$$\sum_a \nabla_\theta \pi_\theta(a \mid s)\, b(s) = 0 \quad \textit{for all } s \in \mathcal{S}.$$

Consequently,

$$\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, \big(Q^\pi(s,a) - b(s)\big)\big].$$

Proof. Since $b(s)$ does not depend on a:

$$\sum_a \nabla_\theta \pi_\theta(a \mid s)\, b(s) = b(s)\, \nabla_\theta \sum_a \pi_\theta(a \mid s) = b(s)\, \nabla_\theta\, 1 = 0.$$

The second statement follows by inserting $Q^\pi(s,a) - b(s)$ in place of $Q^\pi(s,a)$ in the policy gradient theorem and noting that the $b(s)$ term vanishes. □

RL/ML Connection

The most common baseline is $b(s) = V^\pi(s)$, the state-value function. This gives the *advantage function* $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$, and the gradient becomes

$$\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, A^\pi(s,a)\big].$$

Since A^π has zero mean under $a \sim \pi_\theta(\cdot \mid s)$ (by definition), this form typically has lower variance than using Q^π directly. This is the foundation of advantage actor-critic methods.

8 Compatible Function Approximation

In practice, $Q^\pi(s,a)$ is unknown and must be approximated. A natural question is: when does using an approximation $f_w(s,a) \approx Q^\pi(s,a)$ preserve the correctness of the policy gradient? The compatible function approximation theorem gives precise conditions.

Definition 8.1 (Compatible Features). A function approximator $f_w(s,a) = w^\top \nabla_\theta \log \pi_\theta(a \mid s)$ is said to have *compatible features* with the policy π_θ.

Theorem 8.2 (Compatible Function Approximation). *Suppose $f_w(s,a)$ satisfies:*

(i) ***Compatible features:*** $\nabla_w f_w(s,a) = \nabla_\theta \log \pi_\theta(a \mid s)$.

(ii) ***Minimizes projected error:*** *w minimizes the mean squared error $\mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\big[(f_w(s,a) - Q^\pi(s,a))^2\big]$.*

Then

$$\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, f_w(s,a)\big].$$

That is, using f_w in place of Q^π gives the exact policy gradient.

Proof. Condition (ii) implies that the gradient of the MSE with respect to w is zero at the optimal w^*:

$$\mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\big[\nabla_w f_{w^*}(s,a)\,(f_{w^*}(s,a) - Q^\pi(s,a))\big] = 0.$$

Using condition (i), $\nabla_w f_w(s,a) = \nabla_\theta \log \pi_\theta(a \mid s)$, so:

$$\mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\,(f_{w^*}(s,a) - Q^\pi(s,a))\big] = 0.$$

This means

$$\mathbb{E}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, f_{w^*}(s,a)\big] = \mathbb{E}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi(s,a)\big] = \nabla_\theta J(\theta),$$

where the last equality is the policy gradient theorem. □

RL/ML Connection

The compatible function approximation theorem justifies actor-critic architectures. For a linear approximator $f_w(s,a) = w^\top \nabla_\theta \log \pi_\theta(a \mid s)$, condition (i) is automatically satisfied. With a baseline, f_w only needs to approximate the advantage. The natural actor-critic of Peters and Schaal [2008] exploits this by solving for w via least squares, then performing a natural gradient step.

9 Exercises

Exercise 9.1 (Score Function Computation). $(*)$
Consider a Bernoulli policy $\pi_\theta(a = 1 \mid s) = \sigma(\theta^\top \phi(s))$, where $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function. Derive the score function $\nabla_\theta \log \pi_\theta(a \mid s)$ for both $a = 1$ and $a = 0$. Verify that $\mathbb{E}_{a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a \mid s)] = 0$.

Exercise 9.2 (Gaussian Policy with Learnable Variance). $(*)$
Extend Example 3.7 to the case where the variance is also parameterized: $\pi_\theta(a \mid s) = \mathcal{N}(a;\, \mu_\theta(s),\, \sigma_\theta^2(s))$ with $\theta = (\theta_\mu, \theta_\sigma)$. Compute the score function $\nabla_\theta \log \pi_\theta(a \mid s)$ with respect to both θ_μ and θ_σ.

Exercise 9.3 (Unrolling the Recursion). $(**)$
In the proof of Theorem 5.1, verify the unrolling in Step 4 rigorously. Specifically, show by induction that for all $T \geq 1$:

$$\nabla_\theta V^\pi(s) = \sum_{t=0}^{T-1} \gamma^t \sum_{s'} \mathbb{P}^\pi(s \to s', t)\, \psi(s') + \gamma^T \sum_{s'} \mathbb{P}^\pi(s \to s', T)\, \nabla_\theta V^\pi(s'),$$

and prove that the remainder term vanishes as $T \to \infty$.

Exercise 9.4 (Multiple Start States). $(**)$
Suppose the start state is drawn from a distribution μ over $\mathcal{S}$. Show that the policy gradient theorem becomes

$$\nabla_\theta J(\theta) = \sum_s \tilde{d}_\mu^\pi(s) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a),$$

where $\tilde{d}_\mu^\pi(s) = \sum_{t=0}^\infty \gamma^t \sum_{s_0} \mu(s_0)\, \mathbb{P}(S_t = s \mid S_0 = s_0, \pi)$.

Exercise 9.5 (Compatible Approximation for Softmax). $(***)$
For the softmax policy of Example 3.6, explicitly construct the compatible function approximator $f_w(s,a)$. Write out the least-squares problem that determines w^*, and show that f_{w^*} can be interpreted as a linear approximation to the advantage function $A^\pi(s,a)$ in the feature space defined by $\phi(s,a) - \mathbb{E}_\pi[\phi(s,\cdot)]$.

10 Summary

- The **log-derivative trick** ($\nabla_\theta p_\theta(x) = p_\theta(x)\,\nabla_\theta \log p_\theta(x)$) converts the gradient of an expectation into an expectation of a gradient-weighted quantity, enabling Monte Carlo estimation.
- The **score function** $\nabla_\theta \log \pi_\theta(a \mid s)$ has zero mean under $a \sim \pi_\theta(\cdot \mid s)$ and measures how much the log-probability of an action changes with θ.
- The **policy gradient theorem** states that $\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^\pi, a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi(s,a)]$. It does *not* require differentiating through the transition dynamics.
- The theorem holds in both the **discounted episodic** setting (with the discounted state visitation distribution) and the **average-reward** setting (with the stationary distribution and differential action-values).
- **Baseline invariance**: subtracting any state-dependent function $b(s)$ from $Q^\pi(s,a)$ does not change the expected gradient, providing a mechanism for variance reduction.
- The **compatible function approximation** theorem guarantees that a critic with features matching the policy score yields the exact gradient, justifying actor-critic architectures.
- For **softmax policies**, the score is the feature deviation $\phi(s,a) - \mathbb{E}_\pi[\phi(s,\cdot)]$; for **Gaussian policies**, the score is proportional to $(a - \mu_\theta(s)) \cdot \phi(s)/\sigma^2$.

11 Further Reading

- **Lesson 55** develops the REINFORCE algorithm using sampled returns as unbiased estimates of Q^π, and analyzes baseline subtraction for variance reduction.
- **Lesson 56** (Trust Region Methods) constrains policy updates to a KL-divergence neighborhood, leading to TRPO [Schulman et al., 2015].
- Kakade [2002] introduces the *natural policy gradient*, preconditioned by the inverse Fisher information matrix.
- Peters and Schaal [2008] connects compatible features to natural gradients in natural actor-critic methods.
- Bertsekas and Tsitsiklis [1996, Chapter 6] analyzes gradient-based approximate dynamic programming with convergence guarantees.
- Kakade [2003, Chapter 2] treats the regularity conditions for interchanging differentiation and expectation.

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